Deliverable 3

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Imports the

In [1]:
```
import random
from game_simulator_v1 import Character, Warrior, Mage
from items import Weapon
from dice_class import Dice
from battle_class import Battle
```

A presentation of each class

Character classes

First we have the normal caracter class. It start with the arguments name, max_hp, damage and weapon. The name dose not have a default value and must be given. The rest have defualt values. Max_hp default value is of 100, damage is 10 and weapon is none. The character class have the folowing methods: init, attack, heal, introduse, str, repr, hp, take_damage and is_alive. The init method take and give all the arguments to the charachter it self and sets the Hp to be the same as the max_hp in the beginning.

The attack function calls the take_damage function of the target and take damage either from the character damage og weapon. The heal function heals the character by a given amount, and the introduce function prints a introduction of the character. The repr function givs a representation of the character and the str function givs a string representation of the character. The hp function is a read only function that givs the amount of hp the character has left, and the is_alive function checks if the carakter is alive.

The take damage function make the caracter take a given damage and make sure that the hp can't go bellow zero.

In [2]:
```
#Character with default values and no weapon
Per = Character('Per')

print(Per)
print(Per.__repr__)
Per.introduce()

#Character with non defult values and a default weapon
Kåre = Character('Kåre', 110, 11, Weapon('kinfe', Dice()))
print(Kåre)
Kåre.introduce()

#The character attack
```

```
Per.attack(Kåre)
print(Kåre)

Kåre.attack(Per)
print(Per)

#The character heal
Per.heal(2)
print(Per)

Kåre.heal(5)
print(Kåre)

#The hp of a character
print(Per.hp)
```

```
Character: Per with 100 HP and 10 damage and weapon None
<bound method Character.__repr__ of <Character name=Per hp=100 damage=10 weapon=N
one>)>
Hi, my name is Per!
Character: Kåre with 110 HP and 11 damage and weapon <items.Weapon object at 0x00
00025FABB43230>
Hi, my name is Kåre!
Character: Kåre with 100 HP and 11 damage and weapon <items.Weapon object at 0x00
00025FABB43230>
Character: Per with 95 HP and 10 damage and weapon None
Character: Per with 97 HP and 10 damage and weapon None
Character: Kåre with 105 HP and 11 damage and weapon <items.Weapon object at 0x00
00025FABB43230>
97
```

The Warrior and mage classes

The Warrior and mage classes are subclasses / classes that inherets most of the arguments and metods form the character class.

The warrior take one extra argument for the armor and replais the take damage method with a new method that reduses the damage taken by the amount of armor. The default armor is 5.

The mage class have two extra argumenst the mama and max_mana arguments and replaces the attack function with a new functiom that take double damege if the mage have mana and drop the mana by 10, and take normal damage if the mage have too litle mana and gain 5 mana. The default mana and max mana is 30.

In [3]:
```
#Warrior with weapon an defualt armor
Soldier = Warrior('Arne', weapon = Weapon('sword', Dice(10)))

Soldier.introduce()
print(Soldier)

#Mage with weapon and default mana and max mana
Wizard = Warrior('Anne', weapon = Weapon('stav', Dice(10)))

Wizard.introduce()
print(Wizard)
```

```
#The warrior attacks mage
Soldier.attack(Wizard)
print(Wizard)

#The mage atacks warrior
Wizard.attack(Soldier)
print(Soldier)


#The mage attacks multiple times
Wizard.attack(Soldier)
print(Soldier)

Wizard.attack(Soldier)
print(Soldier)

Wizard.attack(Soldier)
print(Soldier)

Wizard.attack(Soldier)
print(Soldier)
```

```
Hi, my name is Arne!
Character: Arne with 100 HP and 10 damage and weapon <items.Weapon object at 0x00
00025FABBA4F50>
Hi, my name is Anne!
Character: Anne with 100 HP and 10 damage and weapon <items.Weapon object at 0x00
00025FABBA5310>
Character: Anne with 96 HP and 10 damage and weapon <items.Weapon object at 0x000
0025FABBA5310>
Character: Arne with 100 HP and 10 damage and weapon <items.Weapon object at 0x00
00025FABBA4F50>
Character: Arne with 100 HP and 10 damage and weapon <items.Weapon object at 0x00
00025FABBA4F50>
Character: Arne with 100 HP and 10 damage and weapon <items.Weapon object at 0x00
00025FABBA4F50>
Character: Arne with 100 HP and 10 damage and weapon <items.Weapon object at 0x00
00025FABBA4F50>
Character: Arne with 100 HP and 10 damage and weapon <items.Weapon object at 0x00
00025FABBA4F50>
```

Dice

Then we have the dice class. It starts with the argument sides. The number of sides has a default value of 6.

init initialize the dice and the number of sides. The number of sides has to be at least 2.

The call method returns a random integer when you roll the dice.

The repr method returns a string representation for exmaple Dice(sides=6)

The call method allows the dice object to be called like a function to roll it.

In [4]:
```
#Dice with default number of sides
dice = Dice()
```

```
print(dice)

print(dice.__repr__)
print(dice.roll())

#We use the call method
print(dice())

#Dice with a chosen number of sides
d20 = Dice(20)

print(d20)

print(dice.__repr__)
print(d20.roll())

#We use the call method
print(d20())
```

```
Dice(sides = 6)
<bound method Dice.__repr__ of Dice(sides = 6)>
1
6
Dice(sides = 20)
<bound method Dice.__repr__ of Dice(sides = 6)>
9
20
```

Create items

The weapon class uses the Dice object to decide the damage. It starts with the arguments name and damage_dice. The name is the name of the weapon for example "The legendary sword". damage_dice is the dice object assosciated with the weapon and calculates the damage from the weapon.

init initializes the weapon with a name and the corresponding dice.

The roll_damage method rolls the weapons damage_dice and returns the damage value.

In [5]:
```
#Create the damage_dice
d15 = Dice(15)

weapon = Weapon("The legendary sword", d15)

print(Weapon)

#We use the weapon and we get the damage value returned.
weapon.roll_damage()
```

```
<class 'items.Weapon'>
```

Out[5]:
```
4
```

Battle class

The battle class take to arguments, it take to caracters. One the hero that atacks first and the enemy that atacks last. In the init metode it thefins the hero and enemy they need to

be character classes. Either Character, Warrior or mage.

The battle class have three methods: the init metode mentened previosly, the playround and simulate batle functions.

The play_round function make the hero atack the enemy and prints the outcome. If the enemy is stil alive the enemy ataks the hero and the outcome gets printed.

The simulate battle function first prints the round number then calls the round function plays a round then it updates the round number. Then it repeats this process till one of the characters are dead and prints the character that won. The function checks if the characters are dead by calling the respective is-alive functions.

The characters play to rounds

In [6]:
```
enemy = Mage('The grang mage')
hero = Warrior('The loyal knight')

enemy.introduce()
hero.introduce()

battle = Battle(hero, enemy)

battle.play_round()

battle.play_round()
```

```
Hi, my name is The grang mage!
Hi, my name is The loyal knight!
The loyal knight attacks for 10 damage without a weapon
Now The grang mage has 90 HP
The grang mage attacks for 20 damage using magic
Now The loyal knight has 85 HP
The loyal knight attacks for 10 damage without a weapon
Now The grang mage has 80 HP
The grang mage attacks for 20 damage using magic
Now The loyal knight has 70 HP
```

The battle without weapons

In [7]:
```
enemy = Mage('The grang mage')
hero = Warrior('The loyal knight')

enemy.introduce()
hero.introduce()

battle_1 = Battle(hero, enemy)

battle_1.simulate_battle()
```

```
Hi, my name is The grang mage!
Hi, my name is The loyal knight!
The battle begins! The loyal knight vs The grang mage
Round 1
The loyal knight attacks for 10 damage without a weapon
Now The grang mage has 90 HP
The grang mage attacks for 20 damage using magic
Now The loyal knight has 85 HP
Round 2
The loyal knight attacks for 10 damage without a weapon
Now The grang mage has 80 HP
The grang mage attacks for 20 damage using magic
Now The loyal knight has 70 HP
Round 3
The loyal knight attacks for 10 damage without a weapon
Now The grang mage has 70 HP
The grang mage attacks for 20 damage using magic
Now The loyal knight has 55 HP
Round 4
The loyal knight attacks for 10 damage without a weapon
Now The grang mage has 60 HP
The grang mage attacks for 10 damage without using weapon
Now The loyal knight has 50 HP
Round 5
The loyal knight attacks for 10 damage without a weapon
Now The grang mage has 50 HP
The grang mage attacks for 10 damage without using weapon
Now The loyal knight has 45 HP
Round 6
The loyal knight attacks for 10 damage without a weapon
Now The grang mage has 40 HP
The grang mage attacks for 20 damage using magic
Now The loyal knight has 30 HP
Round 7
The loyal knight attacks for 10 damage without a weapon
Now The grang mage has 30 HP
The grang mage attacks for 10 damage without using weapon
Now The loyal knight has 25 HP
Round 8
The loyal knight attacks for 10 damage without a weapon
Now The grang mage has 20 HP
The grang mage attacks for 10 damage without using weapon
Now The loyal knight has 20 HP
Round 9
The loyal knight attacks for 10 damage without a weapon
Now The grang mage has 10 HP
The grang mage attacks for 20 damage using magic
Now The loyal knight has 5 HP
Round 10
The loyal knight attacks for 10 damage without a weapon
Now The grang mage has 0 HP
Battle is finished
The loyal knight wins the battle!
```

Battle without weapons and not default values

In [8]:
```
enemy = Mage('The grang mage', 350, 15, 75, 100)
hero = Warrior('The loyal knight', 250, 20, 15)
```

```
enemy.introduce()
hero.introduce()

battle_1 = Battle(hero, enemy)

battle_1.simulate_battle()
```

```
Hi, my name is The grang mage!
Hi, my name is The loyal knight!
The battle begins! The loyal knight vs The grang mage
Round 1
The loyal knight attacks for 20 damage without a weapon
Now The grang mage has 330 HP
The grang mage attacks for 30 damage using magic
Now The loyal knight has 235 HP
Round 2
The loyal knight attacks for 20 damage without a weapon
Now The grang mage has 310 HP
The grang mage attacks for 30 damage using magic
Now The loyal knight has 220 HP
Round 3
The loyal knight attacks for 20 damage without a weapon
Now The grang mage has 290 HP
The grang mage attacks for 30 damage using magic
Now The loyal knight has 205 HP
Round 4
The loyal knight attacks for 20 damage without a weapon
Now The grang mage has 270 HP
The grang mage attacks for 30 damage using magic
Now The loyal knight has 190 HP
Round 5
The loyal knight attacks for 20 damage without a weapon
Now The grang mage has 250 HP
The grang mage attacks for 30 damage using magic
Now The loyal knight has 175 HP
Round 6
The loyal knight attacks for 20 damage without a weapon
Now The grang mage has 230 HP
The grang mage attacks for 30 damage using magic
Now The loyal knight has 160 HP
Round 7
The loyal knight attacks for 20 damage without a weapon
Now The grang mage has 210 HP
The grang mage attacks for 30 damage using magic
Now The loyal knight has 145 HP
Round 8
The loyal knight attacks for 20 damage without a weapon
Now The grang mage has 190 HP
The grang mage attacks for 15 damage without using weapon
Now The loyal knight has 145 HP
Round 9
The loyal knight attacks for 20 damage without a weapon
Now The grang mage has 170 HP
The grang mage attacks for 30 damage using magic
Now The loyal knight has 130 HP
Round 10
The loyal knight attacks for 20 damage without a weapon
Now The grang mage has 150 HP
The grang mage attacks for 15 damage without using weapon
Now The loyal knight has 130 HP
Round 11
The loyal knight attacks for 20 damage without a weapon
Now The grang mage has 130 HP
The grang mage attacks for 15 damage without using weapon
Now The loyal knight has 130 HP
Round 12
The loyal knight attacks for 20 damage without a weapon
```

```
Now The grang mage has 110 HP
The grang mage attacks for 30 damage using magic
Now The loyal knight has 115 HP
Round 13
The loyal knight attacks for 20 damage without a weapon
Now The grang mage has 90 HP
The grang mage attacks for 15 damage without using weapon
Now The loyal knight has 115 HP
Round 14
The loyal knight attacks for 20 damage without a weapon
Now The grang mage has 70 HP
The grang mage attacks for 15 damage without using weapon
Now The loyal knight has 115 HP
Round 15
The loyal knight attacks for 20 damage without a weapon
Now The grang mage has 50 HP
The grang mage attacks for 30 damage using magic
Now The loyal knight has 100 HP
Round 16
The loyal knight attacks for 20 damage without a weapon
Now The grang mage has 30 HP
The grang mage attacks for 15 damage without using weapon
Now The loyal knight has 100 HP
Round 17
The loyal knight attacks for 20 damage without a weapon
Now The grang mage has 10 HP
The grang mage attacks for 15 damage without using weapon
Now The loyal knight has 100 HP
Round 18
The loyal knight attacks for 20 damage without a weapon
Now The grang mage has 0 HP
Battle is finished
The loyal knight wins the battle!
```

The battle with weapons

In [9]:
```
enemy = Mage('The grang mage', weapon = Weapon('The wisest staff', Dice(20)))
hero = Warrior('The loyal knight', weapon = Weapon('The legendary sword', Dice(2

enemy.introduce()
hero.introduce()

battle_2=Battle(hero, enemy)

battle_2.simulate_battle()
```

```
Hi, my name is The grang mage!
Hi, my name is The loyal knight!
The battle begins! The loyal knight vs The grang mage
Round 1
The loyal knight attacks for 2 damage using The legendary sword
Now The grang mage has 98 HP
The grang mage attacks for 20 damage using magic
Now The loyal knight has 85 HP
Round 2
The loyal knight attacks for 8 damage using The legendary sword
Now The grang mage has 90 HP
The grang mage attacks for 20 damage using magic
Now The loyal knight has 70 HP
Round 3
The loyal knight attacks for 14 damage using The legendary sword
Now The grang mage has 76 HP
The grang mage attacks for 20 damage using magic
Now The loyal knight has 55 HP
Round 4
The loyal knight attacks for 12 damage using The legendary sword
Now The grang mage has 64 HP
The grang mage attacks for 10 damage using The wisest staff
Now The loyal knight has 49 HP
Round 5
The loyal knight attacks for 12 damage using The legendary sword
Now The grang mage has 52 HP
The grang mage attacks for 10 damage using The wisest staff
Now The loyal knight has 45 HP
Round 6
The loyal knight attacks for 13 damage using The legendary sword
Now The grang mage has 39 HP
The grang mage attacks for 20 damage using magic
Now The loyal knight has 30 HP
Round 7
The loyal knight attacks for 17 damage using The legendary sword
Now The grang mage has 22 HP
The grang mage attacks for 10 damage using The wisest staff
Now The loyal knight has 25 HP
Round 8
The loyal knight attacks for 15 damage using The legendary sword
Now The grang mage has 7 HP
The grang mage attacks for 10 damage using The wisest staff
Now The loyal knight has 25 HP
Round 9
The loyal knight attacks for 5 damage using The legendary sword
Now The grang mage has 2 HP
The grang mage attacks for 20 damage using magic
Now The loyal knight has 10 HP
Round 10
The loyal knight attacks for 7 damage using The legendary sword
Now The grang mage has 0 HP
Battle is finished
The loyal knight wins the battle!
```